

Sneha Gupta

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Experience

Meril Life Science, AI Engineer

Aug 2024 – Present

- **Stanley AI – Regulatory Intelligence Platform:**
- Built an end-to-end AI system for medical device regulatory automation supporting EU MDR, US FDA, and CDSCO compliance.
- Developed AI services including **risk classification, component recommendation, GSPR generation, and Design History Dossier (DHD)** automation.
- Implemented **multi-agent workflows using LangGraph and LangSmith** with LLMs (OpenAI, Ollama).
- Fine-tuned **Flan-T5 and T5-small models** for domain-specific summarization of regulatory documents, improving contextual accuracy and reducing hallucinations.
- Built **Stella**, a chat-based AI assistant enabling users to generate and manage complete **Design History Dossiers (DHD)** through conversational workflows.
- Developed **Stan**, a multilingual voice AI assistant supporting **Hindi, English, and Gujarati** for querying design inputs, regulatory documents, and project knowledge through voice interactions.
- **Knowledge Graph & Infrastructure:** Built LightrAG-based pipelines by processing FDA data and regulatory PDFs to construct a **Neo4j knowledge graph**.
- Designed scalable architecture using **Milvus (vector DB) + Redis (caching)** and deployed using Docker on on-premise systems.
- **DICOMPixel – Medical Imaging Platform:** Developed microservices for image conversion, metadata extraction, annotation, compression, and segmentation.

Tech Stack: Python, Langchain, Langgraph, Livekit, FastAPI, Hugging Face, LLMs, RAG, Prompt Engineering, Ollama, FAISS, Redis

Education

Btech Pranveer Singh Institute of Technology, Computer Science (AI)

Dec 2020 – June 2024

- GPA: 7.90

Skills

Programming & Frameworks	Python, OpenClaw, Pipecat, LiveKit, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph, Django, FastAPI, SQL
Tools & Libraries	Git, Docker, RESTful APIs, Redis, Sphinx, GitHub, Groq, Postman, CrewAI, Streamlit, Docling, HyperExtract, AWS (EC2, S3, Lambda, SageMaker), PostgreSQL.

Projects

MediScribe AI – Voice Medical Scribe Assistant

- Built an AI voice-scribe pipeline that transcribes and diarizes doctor–patient conversations in real time.
- Designed a multi-agent LLM system (Groq/Llama 3.3) that routes transcripts to parallel agents generating clinical notes, urgency alerts, plain-language summaries, and Ayurvedic assessments (MedGemma/LangGraph).
- Shipped a full-stack product with a FastAPI backend (REST + WebSocket streaming) and a React/Vite dashboard, deployed on Render with GPU/CPU auto-detection.
- **Tools:** Python, Pipecat, FastAPI, WhisperX, pyannote, Groq, LangGraph, MongoDB, React, Docker



Video Captioning and RAG Chatbot

- Built a multimodal video-understanding pipeline that samples video frames, generates scene captions with BLIP, extracts on-screen text with EasyOCR, and transcribes speech from the audio track using WhisperX.
- Orchestrated the pipeline as a LangGraph StateGraph, fusing captions, OCR text, and speech transcripts into a single coherent narrative via an LLM (Groq/Llama 3).
- Built a RAG chatbot (FAISS + sentence-transformers + RetrievalQA) enabling natural-language Q&A over video content, exposed through both a CLI and a Streamlit UI.
- **Tools:** Python, LangGraph, BLIP, EasyOCR, WhisperX, LangChain, Groq/Llama 3, FAISS, Streamlit

